# MICROECONOMICS

# Externalities

**Social efficiency is attained when the marginal social benefit (MSB) of producing any extra good/services exceeds its extra costs to the society which is marginal social cost (MSC).**

- MSB is the additional benefit gained by society of producing or consuming one more unit of a good.
- MSC is the additional cost incurred by society in producing or consuming one more unit of a good.

If MSB>MSC, the production/output should be increased.

If MSB<MSC, the production/output should be reduced,

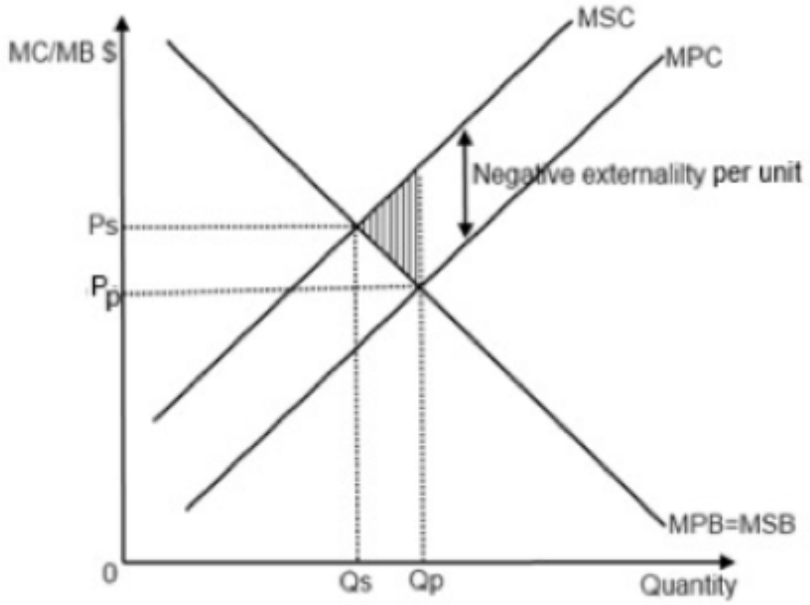
If MSB=MSC, current level of production is optimal (socially efficient level of output).

In reality, MSB=MSC is nearly impossible and it is this inefficiency of the system to allocate resources efficiently which leads to market failures. The governments intervene as when market failures occur.

# Externalities

Marginal Social Cost (MSC) = Marginal Private Cost + External Cost/Externality

Thus the free market equilibrium occurs at the point where Marginal Social Benefit (MSB) equals Marginal Private Cost (MPC). The free market optimum output is $Q_p$. But, the socially optimum level of output is $Q_s$ (where MSB = MSC). The shaded region shows negative externality as more than socially optimum quantity is produced.



# Externalities

**Externalities refers to the costs or benefits enjoyed/incurred by the society in producing or consuming extra units of goods/services.**

Social cost = Private cost + Externalities in production

Social benefit = Private benefit + Externalities in consumption

Externalities can be of different types:

- Positive externalities in production – When a firm provides an external benefit during the production process
- Negative externalities in production – When a firm imposes an external cost during the production process
- Positive externalities in consumption – Social benefit is provided due to the consumption patterns
- Negative externalities in consumption – Social cost is incurred due to the consumption patterns

# Externalities

- When firms that engage in production, pollutes the rivers or environment (external cost), it is a negative externality in production.
- If firms indulge in proper R&D activities, it falls under the category of positive externalities in production.
- When people use their private vehicles more often, it creates lot of pollution and hence is an example of negative externalities in consumption.
- Similarly, when people use mass transport system, it avoid lot of pollution and traffic congestion and can be considered as positive externality in consumption.

**Why does the problem of external cost arise?**

Because nobody has legal ownership of the environment and resources and thus nobody can prevent the harm caused to it. The market for those resources is missing.

# Externalities

What are public goods?

Public goods are a category of goods which has lot externalities involved such that free market does not produce them at all. Two main characteristics of public goods are:

- Non-rivalry: The consumption of a good by one person does not prevent others from consuming the same good.
- Non-excludability: The consumption of such goods is open to all and nobody can be prevented from consuming it.

This results in the free-rider problem.

If a good is available freely for consumption (public goods), people will often be unwilling to pay for it. This free-rider problem arises when nobody can be excluded from the consumption of a good bought by another person. Such goods might have large social benefits (positive externalities) and may be socially desirable. But, they will be privately unprofitable. As a result, we often see that the govt. provides such goods.

# Externalities

**Government Intervention in the Market**

In the case of public goods, when the markets are missing, govt. can interfere to avoid market failure in the following ways:

- Taxes and subsidies
- When market imperfections occur, the output produced will either be more than or less than the socially optimum level of output. The golden rule is to tax the goods which are produced more than the optimum level and to subsidise the goods which are produced less than the optimum.
- Legislation and regulation

Legal restrictions are easy to implement and administer. The laws can be mainly of three types: those that prohibit or regulate behaviour that imposes external costs; those that prevent firms from providing false/misleading information; and those that prevent/regulate the monopolies and oligopolies.

# Externalities

- Government provisions

When it comes to the public goods and merit goods, which the private companies do not provide or find unprofitable, government should take the responsibility of its provision. This ensures the right amount of consumption of such goods. Another way in which govt. can intervene is by providing proper information regarding harmful goods/services.

How to choose the proper tax amount for a good causing negative externality?

Tax amount should be such that it should be the difference between the Marginal Social Cost (MSC) and Marginal Social Benefit (MSB) at the optimum level of output. How will you illustrate this graphically?

# THANK YOU